

Anirudh Rao

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EDUCATION

2021–2026 **Indian Institute of Technology Madras (IIT Madras)**
B. Tech in Biological Engineering and Integrated M. Tech in Data Science
CGPA of 9.73

ACADEMIC ACCOMPLISHMENTS

2025 Won the **Institute Merit Prize** for having the best academic performance in the third year of the Biological Engineering programme.

2024 Won the **Dr. Anita Mehta-Damani Prize** for having the best academic performance in the second year of the Biological Engineering programme.

2024 Selected as a **Khorana Scholar** to undertake research activities in the fields of biology and biotechnology at an academic institution in the United States.

2023 Selected as a **Young Research Fellow** to participate in undergraduate research at IIT Madras.

2020 Selected for the **Kishore Vaigyanik Protsahan Yojana (KVPY)**, a prestigious science fellowship in India.

RESEARCH EXPERIENCE

2025–PRESENT **Baker Lab, University of Washington**
Advisor: Dr. David Baker
Working on improving the design success of *de novo* protein binders using deep learning and large-scale experimental data.

2024 **Cong Lab, Stanford University**
Advisor: Dr. Le Cong
Worked on *in silico* optimisation of ADAR2-Cas13-mediated RNA editing in mammalian cells using protein language models.
Developed a computational pipeline to automate the analysis of killing assays to test the efficacy of NK cell-mediated cancer immunotherapy. Performed digital cytometry and survival analysis using publicly available data from TCGA.

2023–2024 **Computational Systems Biology Lab, IIT Madras**
Advisor: Dr. Karthik Raman
Worked on designing and implementing novel approaches and applications of biomolecular computing to solve computationally challenging problems.
Designed and experimentally validated an *in vitro* artificial neural network for universal function approximation using DNA molecules.

2022–2025

iGEM IIT Madras

Team Lead (2024–2025): Developed a co-culture of engineered silicon-solubilizing bacteria and diatoms for *in situ* resource utilization during Mars missions. Created Astrolabe, the first software tool for chassis recommendation in space synthetic biology (**Gold Medal**, iGEM 2024).

Lab Head (2023–2024): Constructed and characterized a library of synthetic ribosome binding sites (RBS) for use in recombinant *Lactococcus lactis*. Built a machine learning model for prediction and optimization of gene expression by RBS engineering (**Gold Medal**, iGEM 2023).

Lab Member (2022–2023): Designed protocols for the directed evolution of yeast cytosine deaminase to enhance stability for enzyme prodrug cancer therapy (**Scientific Contribution Award**, iDEC 2022). Constructed genetic circuits in *E. coli* to produce bacterial cellulose from whey water for transdermal insulin patches (**Grand Prize**, iGEM Indian League 2022).

PREPRINTS AND PUBLICATIONS

Dinesh, R., Wang, X., Mohammad, I., Gunasekaran, P., Stikloraitis, K., Villafuerte, J., **Rao, A.**, Hernandez-Lopez, R., Sunwoo, J., & Cong, L. (2025). Surfaceome CRISPR activation screening uncovers ligands regulating tumor sensitivity to NK cell killing. *bioRxiv*. doi: 10.1101/2025.09.17.676710.

SKILLS

PROGRAMMING AND ML	Python, Java, C, MATLAB; ML with TensorFlow, Keras, PyTorch, Scikit-Learn; MLOps with Airflow, Spark, Docker, Git, FastAPI, DVC, Prometheus, Grafana, MLflow
COMPUTATIONAL BIOLOGY	Molecular dynamics (VMD, GROMACS); COBRA flux balance analysis; RNA-Seq in R (Seurat); network modelling (Cytoscape, NetworkX); <i>de novo</i> protein design (RFdiffusion, ProteinMPNN)
MOLECULAR BIOLOGY	Plasmid isolation, agarose gel electrophoresis, cloning, Gibson assembly, PCR, qPCR, heat-shock transformation, SDS-PAGE

RELEVANT COURSEWORK

Molecular Biology; Structural Biology; Genomics and Proteomics; Synthetic Biology; Data Structures and Algorithms for Biology; Bioinformatics; Computational Systems Biology; Computer Simulations of Biomolecular Systems; Analysis and Interpretation of Biological Data; Foundations of Machine Learning; Machine Learning Operations.

Co-CURRICULAR ACTIVITIES

2023–2024	Teaching Assistant for Dr. K. Subramaniam's course on Biochemistry as part of NPTEL, the National Programme on Technology Enhanced Learning
2023–2024	Social Secretary of Alakananda Hostel, IIT Madras
2023–2024	Student Mentor in the Student Mentorship Cell, IIT Madras
2022–2023	Coordinator, Quiz Club and Word Games Club, IIT Madras
2022–2024	Member, Cultural Contingent, IIT Madras